KEPLER

Planetary Image Processing Suite

User Guide

Version 1.1.5

Introduction

Kepler is a planetary image processing application designed for amateur and professional astronomers. It provides multi-scale wavelet sharpening, FFT frequency-domain denoising, per-channel color correction, limb-artefact removal, and batch processing — all in a single, non-destructive workflow.

Recommended Workflow

Kepler's processing tabs are designed to be applied in order from left to right. Each tab feeds into the next:

Setting	Range	Default	Description
1	Wavelet	—	Sharpen planetary detail at multiple spatial scales
2	FFT	—	Remove high-frequency noise in the frequency domain
3	RGB	—	Correct color balance, gamma, and saturation
4	De-rind	—	Suppress limb sharpening artefacts and onion-ring patterns
5	Batch	—	Apply current settings to multiple files and create animations

Tip: You do not have to use every tab. A typical session uses Wavelet and RGB only. Enable FFT and De-rind only when needed.

Opening and Saving Images

Open Image: Click the Open Image button in the top-left panel. Kepler supports TIFF (8-bit and 16-bit), PNG, and JPEG. For best results use 16-bit TIFF from your stacking software.

Export Image: Click Export to open a small format picker dialog — choose TIFF 16-bit, PNG, JPEG, or BMP. After selecting the format, a file save dialog opens with the correct extension pre-filled. TIFF 16-bit preserves full dynamic range via tifffile. If a preview rotation is set, the rotation is included in the saved file.

Reset: The Reset Layout button restores the default panel sizes. Right-click or double-click any slider to reset it to its default value. Reset Tab Sliders resets all sliders on the current tab. Reset All resets every slider in the application.

Tools Tab

The Tools tab sits between the RGB and De-rind tabs in the pipeline. It applies Levels & Curves followed by CLAHE, after color adjustment and before de-rinding.

Levels & Curves

Per-channel tonal control: black point, white point, gamma, and a draggable tone curve.

Setting	Range	Default	Description
Channel	All / R / G / B	All	All applies identical settings to every channel. Switch to R, G, or B for per-channel adjustments.
Black point	0.00–0.99	0.00	Input values below this are clipped to black. Raises the shadow floor and increases contrast in the bright regions.
White point	0.01–1.00	1.00	Input values above this are clipped to white. Lower to stretch and brighten the image.
Gamma	0.1–4.0	1.00	Power-law tone curve. Values >1 brighten midtones; <1 darken them.

The Tone Curve canvas (300×300px) shows the image histogram in the background and the current spline curve. Drag the anchor circles at the bottom-left (shadows) and top-right (highlights) to reshape the curve. Click anywhere on the curve to add a new control point; right-click a point to remove it.

Tip: For a planetary image that looks flat and dark, try lowering the White point to 0.2–0.4 to stretch the brightness range, then use the tone curve to add a subtle S-curve for contrast.

Reset Curve resets the current channel's curve to the identity line. Reset All resets all channels and all level spinboxes to defaults.

CLAHE

Contrast Limited Adaptive Histogram Equalisation — enhances local contrast across the image independently in small tiles, making subtle planetary features more visible without blowing out bright areas.

Setting	Range	Default	Description
Enable CLAHE	On / Off	Off	CLAHE is disabled by default. Adjusting any spinbox auto-enables it.
Clip Limit	0.5–10.0	0.5	Controls how aggressively contrast is enhanced per tile. Higher values give stronger enhancement but can amplify noise.
Tile Grid	4–64	4	Number of NxN tiles. Smaller values give more localised contrast enhancement.

Setting	Range	Default	Description
Strength	0.00–1.00	0.05	Blends the CLAHE result with the original. 0 = no effect, 1 = full CLAHE. A low value like 0.05 gives a subtle but effective boost.
Apply to	Luminance only / RGB channels	Luminance only	Luminance only is recommended — it enhances contrast without shifting colors. RGB mode can cause color casts.

Tip: For most planetary images, start with the defaults (Clip Limit 0.5, Tile Grid 4, Strength 0.05) and increase Strength gradually until belt and zone detail becomes more visible.

Preview and Magnifier

The left panel contains a side-by-side preview (original left, processed right), a magnifier, and a histogram.

- Zoom: Use the zoom controls or scroll wheel over the preview to zoom in and inspect sharpening quality at pixel level.
- Magnifier: Shows a fixed-position detail crop updated in real time as you adjust sliders. Click the processed canvas to move the magnifier target.
- Histogram: Displays the luminance histogram of the processed image. Watch for clipping at the left (black crush) or right (white blow-out) edges.
- Double-click either preview canvas to open a full-resolution zoom window.

Preview Rotation

A Rotate row sits just below the zoom controls. It lets you rotate the processed image directly from the Preview section without going to the Batch tab.

Setting	Range	Default	Description
Spinbox (−360 – +360°)	Any angle	0°	Type any rotation angle or use the up/down arrows. Positive = clockwise.
−90° / 90° / 180°	Preset buttons	—	One-click rotation to the most common planetary orientations.
⊘ Reset	Button	—	Returns the rotation to 0° (no rotation).

The rotation is applied instantly to the PROCESSED canvas without re-running the pipeline. It persists through all zoom and pan operations.

When you export the image, the current rotation is included in the saved file — both TIFF 16-bit and PNG/JPEG/BMP exports apply the rotation.

Opening a new image resets the rotation to 0°.

Tip: The preview rotation and the Batch tab's ALIGNMENT & ROTATION section stay in sync. Setting a rotation in the Preview automatically ticks Rotate all images in the Batch tab and sets the same angle — so if you then run a batch, the same rotation is applied to every file. Changing the angle in the Batch tab also updates the Preview rotation spinbox.

Wavelet Tab

The Wavelet tab is the core of Kepler's processing pipeline. It decomposes the image into multiple spatial frequency layers, applies independent sharpening and denoising to each, then reconstructs the result.

How Wavelet Sharpening Works

Kepler uses multi-scale unsharp masking (USM). For each layer, a smoothed version of the image is subtracted from the original to extract fine detail at that scale. The detail is amplified and added back. Three layers are processed simultaneously, each targeting a different spatial frequency range.

Layer 1 targets the finest detail — sub-pixel grain and the sharpest edges. Default Decomp σ: 0.5 px.

Layer 2 targets mid-scale structure — belt boundaries, cloud bands. Default Decomp σ: 1.0 px.

Layer 3 targets coarse structure — large-scale contrast across the disk. Default Decomp σ: 2.0 px.

Note: These sigma defaults (0.5 / 1.0 / 2.0 px) are the calibrated sub-pixel kernel operating points for optimal planetary sharpening.

Layer Controls

Each of the three layers has its own card with three sliders and a SharpenFilter:

Setting	Range	Default	Description
Sharpen	0–200	0	Amplification of the extracted detail. 0 = no sharpening. 200 = maximum. A typical planetary starting point is 150–200 on Layer 1, 80–120 on Layer 2, 40–80 on Layer 3.
SharpenFilter	0.0–2.0	0.146 / 0.100 / 0.000	Adds a Difference-of-Gaussians (DoG) component to the detail kernel, emphasising edges within each layer's frequency band. The defaults are the Kepler-calibrated values (0.146 / 0.100 / 0.000).
Bilateral checkbox	On / Off	On	Visible only when Bilateral filter is selected. Uncheck to make that layer use Gaussian instead — useful for keeping fine L1 detail as standard Gaussian while applying bilateral edge-preservation to L2 and L3 only.

Note: Decomp σ and Denoise (noise gate) sliders are hidden from the UI but remain wired to the pipeline. Autosize Filter and Estimate Filter still update them silently.

Autosize Filter

The Autosize Filter button sets Decomp σ and SharpenFilter for all three layers to the calibrated defaults (σ = 0.5 / 1.0 / 2.0 px, SharpenFilter = 0.146 / 0.100 / 0.000). Use this to reset layers to the reference starting point after manual adjustments.

Bilateral Filter Controls

These controls appear in the SHARPEN FILTER card only when bilateral is selected as the filter type.

Setting	Range	Default	Description
Radius	1–10	2	Controls how strongly edges are preserved. Radius 10 = output is identical to Gaussian at the same slider values. Radius 1 = strong edge/limb preservation — flat disk interior gets full sharpening, planet limb and belt edges are barely sharpened, giving a subtle softening effect at those boundaries.
Per-layer checkbox (in each layer card)	On / Off	On	When bilateral is selected, each layer card shows a Use bilateral on this layer checkbox. Uncheck to use Gaussian for that layer. Useful when you want bilateral edge preservation on L2 and L3 but standard Gaussian on L1.

Tip: The bilateral filter is most useful for suppressing sharpening artefacts at the planetary limb while preserving interior detail. Start at Radius 3–5 for a balanced result.

Z-Gaussian Filter Controls

This control appears in the SHARPEN FILTER card only when zgaussian (Z-Gaussian) is selected.

Setting	Range	Default	Description
Factor	0.0–1.0	1.0	Controls the strength of the Z-Gaussian effect. Factor 0 = full Z-Gaussian — stronger and crispier than Gaussian, but you only need about 30% of the Gaussian slider values for a similar visual impact (e.g. slider 67 ≈ Gaussian slider 200). Factor 1 = identical to Gaussian, same slider values for the same result. Intermediate values blend smoothly between the two.

Note: When using Z-Gaussian with Factor 0, start with Sharpen values around 60–70 on Layer 1 rather than the usual 150–200. The filter is inherently stronger and will over-sharpen at high slider values.

Global Controls

Setting	Range	Default	Description
Pre-Smooth σ	0.0–5.0 px	0.0 (off)	Applies a Gaussian blur before wavelet decomposition. Removes sub-pixel noise that would otherwise be amplified by Layer 1. Use 0.5–1.5 on noisy images.
Color Model	oklab / hsl / hsv	oklab	The color space in which sharpening is performed. Oklab is perceptually uniform and avoids color fringing. Recommended: leave as oklab.
Filter	gaussian / zgaussian / bilateral / b3	gaussian	The smoothing kernel for wavelet decomposition. Gaussian is the recommended default. Selecting bilateral or zgaussian reveals extra controls described in the sections below.
Convolve	L / RGB / LRGB	LRGB	Which image channels are sharpened. LRGB performs a luminance pass then a per-channel RGB pass — produces the strongest result. L is the safest (luminance only, no color change).
Power Fn	On / Off + exponent	Off	Applies a power-law non-linearity to the sharpening amplitudes, reducing correction on very bright or very dark pixels. Leave off for most planetary work.

FFT Tab

The FFT tab applies a frequency-domain low-pass filter to reduce high-frequency noise introduced by sharpening. It is disabled by default and should be enabled when the image shows grain or ringing after wavelet sharpening.

Controls

Setting	Range	Default	Description
Enable	On / Off	Off	Activates FFT denoising. When off, the tab has no effect.
Layer	PRE1/PRE2/PRE3 / POST	POST	PRE inserts the filter before wavelet sharpening on the specified layer. POST applies after all sharpening. POST is the typical choice.

Setting	Range	Default	Description
Start %	0–100	60	The frequency (as a percentage of Nyquist) at which attenuation begins. Lower values = more aggressive filtering.
End %	0–100	90	The frequency at which the filter reaches full attenuation (zero). The rolloff ramp runs from Start to End.
Curve	0–100 slider	50 (linear)	Shape of the rolloff. 0 = gentle concave-down curve. 50 = linear. 100 = aggressive concave-up curve. Higher values cut more aggressively.
Auto Denoise	Button	—	Analyzes the power spectrum of the processed image and estimates appropriate Start and End values. Use as a starting point then fine-tune manually.

Tip: The FFT tab shows a live power spectrum and filter curve overlay. The filter start should sit just above the signal content, cutting only into the noise floor.

Auto Denoise

Auto Denoise works by fitting a 1/f power-law curve to the low-frequency signal region of the power spectrum, then finding where the actual spectrum diverges from that curve — the noise floor onset. Start is set to that point (clamped to 15–45% Nyquist), End is fixed at 70%. It operates on the working image (post-wavelet), so run the wavelet first.

Note: Auto Denoise gives a close starting point, not a final answer. It is intended to get you near the right values so you can fine-tune manually.

RGB Tab

The RGB tab provides per-channel color correction, gamma adjustment, and global image controls. Applied after wavelet sharpening and FFT denoising.

Per-Channel Controls

Each of the R, G, and B channels has three independent controls:

Setting	Range	Default	Description
Gain % (0–100)	0–100 displayed (0–200 internal)	50	Multiplies channel brightness. 50 displayed = unity (no change). Above 50 brightens, below 50 darkens. Internally the slider runs 0–200; the displayed value is halved for readability.

Setting	Range	Default	Description
Gamma ÷100 (0.5–2.0)	50–200 (shows 0.50–2.00)	100 (1.00)	Power-law tone curve. 100 = gamma 1.0, no change. Above 100 = midtones brightened. Below 100 = midtones darkened.
Black Point	0–100 %	0	Lifts the shadow floor. Pixels at or below this level map to black; the rest of the range is stretched.

Global Controls

Setting	Range	Default	Description
Saturation (0–100)	0–100 displayed	50	Scales color saturation uniformly. 50 = no change. 0 = grayscale. Apply carefully — over-saturation on planetary data creates false color fringing.
Vibrance (0–100)	0–100 displayed	50	Boosts saturation selectively on less-saturated pixels while leaving already-vivid colors largely unchanged. Gentler than Saturation.
Brightness	Slider	0	Additive brightness offset applied uniformly to all channels after gain.
Contrast	Slider	1.00	Scales pixel values around the midpoint (0.5).
Hue Rotation	−180–+180°	0	Rotates all colors around the hue wheel. Rarely needed for planetary work.

Tip: The Gain, Saturation and Vibrance sliders display 0–100 but are internally scaled 0–200. The neutral (no-change) point is at 50 displayed / 100 internal. Slider default = 50 displayed.

Auto RGB Balance

Analyzes the image histogram and sets Gain and Black Point for each channel automatically. Green is used as the reference channel. The result equalises the shadow and highlight edges of R, G, B so the histogram channels align.

Tip: Auto RGB Balance is a good starting point but rarely produces a perfect result. Use it to get close, then fine-tune Gain and Black Point manually.

De-rind Tab

The De-rind tab suppresses the halo and onion-rind artefacts that appear around bright planetary limbs after aggressive wavelet sharpening. It is disabled by default.

Controls

Setting	Range	Default	Description
Enable De-rind	On / Off	Off	Activates the De-rind algorithm. When off, the tab has no effect.
Edge (outer extent, px)	1–40	8	How far outward from the detected limb the mask extends into the sky. Set to approximately the visible width of the halo in the image.
Feather (inward falloff, px)	0–30	4	How far inward from the limb the mask tapers. A larger feather catches ringing that extends into the disk interior.
Smooth (mask breadth, px)	0–20	1	Gaussian sigma applied to the mask to smooth its edges and prevent hard transitions.
Inset (pull mask inward, px)	0–60	0	Shifts the entire mask band inward from the limb. At Inset = Edge, the sky side of the mask disappears entirely and the band sits inside the disk. Useful for targeting internal ringing patterns.
Auto De-rind	Button	—	Estimates Edge and Feather values from the image's brightness profile near the limb.
Auto Edge	Button	—	Estimates the Edge value only by finding where sky brightness drops to background level.

Gap Controls

Restricts the mask to exclude part of the ring arc. Used for Saturn to exclude the rings from De-rind correction.

Setting	Range	Default	Description
Gap Width (°)	0–180°	0	Angular width of the gap cut into the mask. 0 = no gap.
Gap Angle (°)	0–359°	0	Center angle of the gap (0° = top, clockwise).
Saturn mode	On / Off	Off	Automatically applies symmetric 180° gaps on both sides of the disk, suitable for Saturn's rings.

Show De-rind Mask Overlay

When checked, the processed preview shows the De-rind mask as a red tint overlay. The red band indicates which pixels are being corrected. The planet interior and sky outside the band are not tinted. Use this to verify mask placement before applying.

> **Tip:** If you see two concentric red rings, the Dark Edge feature in Advanced is extending the mask outward. This is intentional — it catches diffraction rings beyond the main halo band.

Advanced Controls

Setting	Range	Default	Description
Dark Edge	On / Off	On	Extends the mask outward to suppress bright diffraction rings beyond the main halo band. Particularly useful on Mars. The extended zone blends into the main mask as a single continuous band.
Pre-blur σ	0–5	0.0 (off)	Blurs the image before De-rind processing. Useful when ringing is very strong.

Disk Detection

All De-rind functions (mask, Auto De-rind, Auto Edge) detect the planet disk using a threshold of 15% of the image peak brightness. This is robust for any disk size — from a full-frame Moon to a small Mars — and always sits well above sky noise (which is typically below 1% of peak).

Batch Tab

The Batch tab applies the current Wavelet / FFT / RGB / De-rind settings to multiple files and optionally creates an animated WebP or GIF from the results.

Workflow

The intended workflow is:

- Open one representative image from your batch set using the normal Open Image button.
- Tune your settings in the Wavelet, FFT, RGB, and De-rind tabs as normal — you are editing that one image and watching the preview.
- Go to the Batch tab. Click Refresh Summary to confirm the settings that will be applied.
- Add files to the INPUT FILES list.
- Set the output folder and folder name.
- Click Run Batch.

Note: The reference image you tuned can be included in the batch file list or excluded — Kepler applies the same settings regardless. Including it will re-process and save it with the _KEP suffix alongside the others.

Input Files

Setting	Range	Default	Description
＋ Add	Button	—	Opens a file picker to add images to the batch list. Supports TIFF, PNG, JPEG, BMP.

Setting	Range	Default	Description
⨯ Remove	Button	—	Removes the selected file(s) from the list.
⊘ Clear	Button	—	Clears all files from the list.

Alignment & Rotation

Both controls are optional and independent.

Setting	Range	Default	Description
Align planet disks	On / Off	Off	Detects the disk center in each image and shifts it to match the first file in the list. Uses the same 15%-of-peak disk detection as De-rind.
Rotate all images	On / Off	Off	Rotates every processed image by the specified angle. Positive = clockwise, negative = counter-clockwise.
Presets	45° / 90° / 180° / -45° / -90°	—	Sets the rotation angle to a common value with one click.
Custom angle (°)	Spinbox −360–+360	0.0	Type any rotation angle including fractional degrees.
Expand canvas	On / Off	Off	When on, the output image is enlarged to contain the full rotated result without cropping. May change image dimensions.

Rotation Preview

A live 180×180 thumbnail in the ALIGNMENT & ROTATION card updates automatically whenever the angle, preset, or Expand canvas setting changes. The preview renders from the current processed image so it reflects the full pipeline. Below the preview, a caption shows the angle, direction (CW/CCW), and if Expand is on, the output dimensions.

Output

Setting	Range	Default	Description
Save to	Folder picker	—	The parent folder where the output folder will be created.
Folder name	Text field	—	Name of the output folder created inside Save to. All processed files are saved here.
Format	TIFF 16-bit / PNG 8-bit / JPEG	TIFF 16-bit	File format for saved images. TIFF 16-bit preserves full bit depth.

Files are named originalfilename_KEP.ext — the _KEP suffix is fixed. The full output path is shown as a live preview below the Folder name field as you type.

Animation

Creates an animated WebP or GIF from the batch processed files automatically after Run Batch completes, or on demand from already-processed files using Export Now.

Setting	Range	Default	Description
Create animation	On / Off	Off	When on, the animation is built automatically after batch processing finishes.
Format	WebP / GIF	WebP	WebP supports full color at quality 90. GIF is color-quantised to 256 colors but more universally supported.
Delay	0.05–10.0 s	0.50	Duration of each frame in seconds.
Loop	On / Off	On	Whether the animation loops continuously.
Filename	Text field	kepler_animation	Output filename (without extension) saved in the same output folder.
▶ Export Now	Button	—	Builds the animation immediately from existing files in the output folder, without re-running the batch.

Settings Summary

The SETTINGS SUMMARY card shows a read-only snapshot of all four pipeline tabs. Click Refresh Summary to update it. Review this before clicking Run Batch to confirm the correct settings will be applied.

Run

Setting	Range	Default	Description
▶ Run Batch	Button	—	Starts batch processing in a background thread. The UI remains responsive.
■ Cancel	Button	—	Stops processing after the current file completes.

A progress bar, per-file status line, and scrollable log track progress. The log shows alignment shifts, rotation applied, files saved, and any errors.

Stats Tab

The Stats tab displays metrics about the current processed image: pixel value statistics, sharpness estimates, and histogram data. Useful for comparing different processing settings quantitatively.

Practical Tips

Starting Settings for Planetary Work

Jupiter

A typical Kepler session for a well-stacked Jupiter image:

- Wavelet tab: Set Layer 1 Sharpen to 150–200, Layer 2 to 80–120, Layer 3 to 40–80. Convolve = LRGB. SharpenFilter defaults are already set correctly by Autosize.
- FFT tab: Leave disabled unless you see grain. If needed, enable POST and click Auto Denoise to get a starting estimate, then adjust Start and End manually.
- RGB tab: Click Auto RGB Balance to neutralize color casts. Fine-tune Gain and Black Point per channel. Vibrance 55–65 helps bring out subtle belt color.
- De-rind tab: Enable if a halo ring is visible around the limb after sharpening. Click Auto De-rind first. Enable Show Mask overlay to verify placement.
- Tools tab: Optional. Try CLAHE (defaults: Clip 0.5, Grid 4, Strength 0.05) to bring out low-contrast belt detail. Use the tone curve to add a gentle S-curve for contrast.

Saturn

Saturn presents a wide, low-contrast disk with fine ring detail and subtle atmospheric banding. The rings are typically much brighter than the disk:

- Wavelet tab: Layer 1 Sharpen 130–160 (blue/UV channels are noisier — keep lower than Jupiter). Layer 2 to 70–90 for ring structure and belt boundaries. Layer 3 to 30–50 for large disk contrast. Convolve = LRGB.
- FFT tab: More useful than for Jupiter — Saturn stacks often have residual seeing noise. Enable POST, set Start to 25–35% and click Auto Denoise. The rings can show aliasing artefacts at high sharpening so FFT denoising helps.
- RGB tab: Saturn's color is subtle — pale yellow/gold disk, off-white rings. Use Auto RGB Balance then reduce Blue Gain slightly (90–95) to warm the tone. Vibrance 40–55. Keep Saturation conservative to avoid unnatural color.
- De-rind tab: Essential for Saturn — the ring-disk junction creates a strong limb-like edge. Click Auto De-rind, then check the mask carefully with Show Mask overlay. The Saturn mode checkbox helps handle the ring gap geometry.
- Tools tab: CLAHE at low strength (0.03–0.05) can reveal subtle Cassini Division detail and equatorial band structure that is otherwise invisible.

Mars

Mars is a small disk with strong albedo contrast between bright deserts and dark regions. The disk is rarely large enough to benefit from Layer 3 sharpening:

- Wavelet tab: Layer 1 Sharpen 150–180 for surface detail — dust storm edges and polar cap boundary sharpen well. Layer 2 to 80–100 for albedo feature boundaries. Layer 3 to 20–40 only — keep low, the disk is too small for large-scale sharpening to be useful. Convolve = LRGB.
- FFT tab: Mars stacks at low altitude can have significant atmospheric dispersion noise. Enable POST and use Auto Denoise as a starting point. The small disk means the FFT operates mostly on sky pixels — verify the power spectrum shows a clear signal peak before trusting the estimate.
- RGB tab: Mars color ranges from pale pink to deep orange depending on dust loading. Use Auto RGB Balance, then increase Red Gain to 105–115 and reduce Blue Gain to 85–95 to bring out the characteristic red. Vibrance 50–60. Black Point 5–10 on all channels to crush the dark background.
- De-rind tab: Less critical than Jupiter or Saturn but still useful if sharpening produces a glow at the limb. Keep Edge low (30–50) and Inset 0–5% to avoid cutting into the polar cap.
- Tools tab: The tone curve is particularly useful for Mars — lift the shadows slightly to reveal dark albedo features, then compress the highlights to prevent the bright polar cap from blowing out.

Avoiding Over-Sharpening

- Zoom to 100% in the preview before evaluating sharpness. Downscaled previews hide ringing.
- If the limb shows a concentric halo ring, reduce Layer 1 Sharpen or enable De-rind.
- If fine surface texture looks grainy, increase Layer 1 Threshold to 10–20.
- If color fringing appears on belt edges, switch Convolve from LRGB to L.

Batch Processing Tips

- Process one representative image first and confirm it looks right before running the batch.
- Use Refresh Summary in the Batch tab to confirm your settings before Run Batch.
- Enable Align planet disks if your images were taken at different orientations or the seeing varied during the session.
- Use Rotation Preview — the live thumbnail — to confirm the rotation angle before running.

Slider Reset

Right-click or double-click any slider to reset it to its default value. Use Reset Tab Sliders to reset all sliders on the current tab. Use Reset All to reset every slider in the application.

Supported Image Formats

- 16-bit TIFF: Full precision preserved. Recommended for input and output.

- 8-bit TIFF / PNG / JPEG: Supported for input. JPEG is not recommended due to compression artefacts that interfere with sharpening.
- Batch output: TIFF 16-bit, PNG 8-bit, or JPEG. TIFF is recommended for archival use.